Cognitive-AI Approaches to Adaptive Scheduling in Braille Memory Retention Using Hybrid SM-2 Models

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researchMethodology = [Jonathan, Benedictus, Christopher]
artificialIntelligence = [Edbert, Jonathan]
```

The Problem

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90%

of legally blind individuals in the US never achieve Braille fluency

Current learning methods are inefficient and unsustainable

Spaced Repetition: The Science

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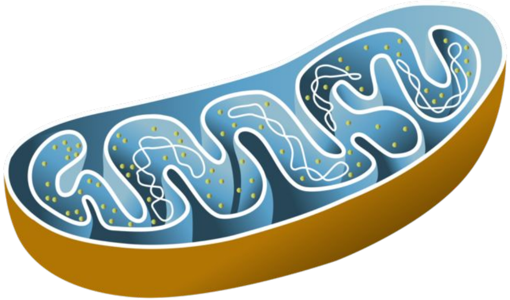
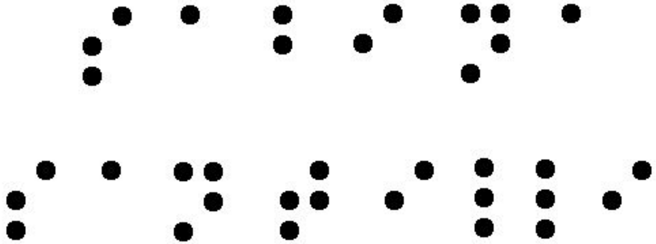
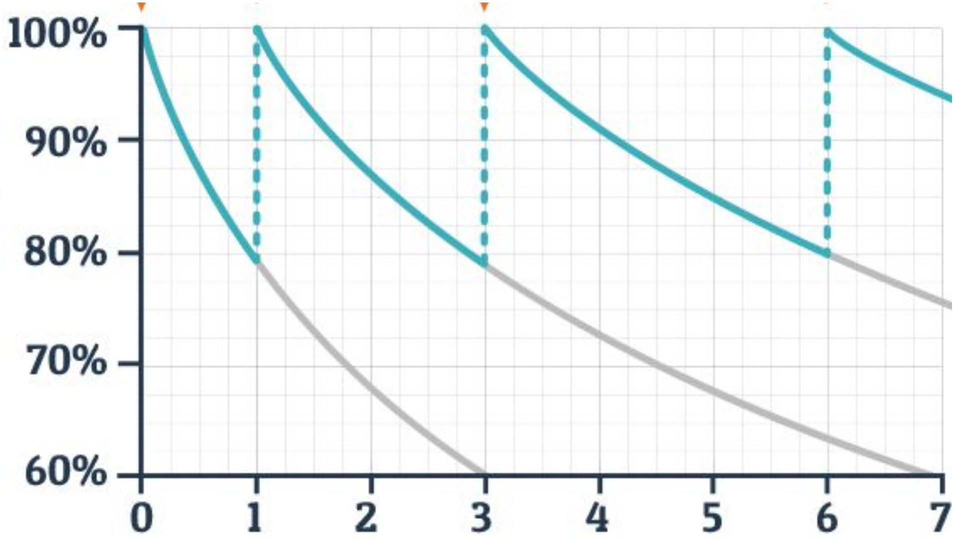




The Forgetting Curve (Ebbinghaus, 1885)

- **60%** retention after 1 day
- **70-75%** after several days
- Continues declining without intervention



Spaced Repetition Resets the Curve

- Resets the forgetting curve at optimal moments
- Strengthens long-term memory consolidation
- Requires precise timing too early wastes time, too late loses the memory
- 96% of studies show spacing benefits

Existing Approaches

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SM-2 Algorithm (SuperMemo[Anki])

Pros:

- Transparent and interpretable
- Simple ease factor updates
- Low computational cost

Cons:

- Fixed rules, no personalization
- Can't adapt to individual learning patterns
- One-size-fits-all approach

SM-2 Algorithm

Ease Factor Calculation

$$EF' = EF + [0.1 - (5-q)(0.08 + 0.02(5-q))]$$

- EF = Ease Factor (difficulty rating)
- q = Quality score (0-5, where 5 = perfect recall)
- $EF' \geq 1.3$ (minimum bound)

SM-2 Algorithm

Interval Scheduling

$I_1 = 1$ day

$I_2 = 6$ days

$I_n = I_{n-1} \times EF'$ (for $n > 2$)

Machine Learning
(FSRS[RemNote], HLR[duolingo])

Pros:

- Highly adaptive and personalized
- Learns from individual patterns
- Superior short-term accuracy
- 20-30% reduction in reviews per day

Cons:

- Black box no transparency
- High computational demands
- Risk of overfitting

Machine Learning
(HLR[duolingo])

$$p = 2^{-\Delta/h},$$

$$\hat{h}_\Theta = 2^{\Theta \cdot x},$$

- p = Recall probability (0 to 1) - chance you'll remember right now
- Δ (delta) = Time elapsed since last review (in days)
- h = Memory half-life (days until p drops to 0.5)
- ĥ (h-hat) = Predicted half-life from the model
- Θ (theta) = Weight vector [θ_0, θ_1, θ_2] - learned from data
- x = Feature vector [bias=1, √(1+n_right), √(1+n_wrong)]
  - n_right = number of correct recalls
  - n_wrong = number of failed recalls

How HLR Works

$$p = 2^{-\Delta/h},$$

$$\hat{h}_\Theta = 2^{\Theta \cdot x},$$

- Learns weights (Θ) from millions of reviews
- Features: √(correct count), √(error count), constant bias term
- More successes → higher √(n_right) → longer predicted half-life
- More failures → higher √(n_wrong) → shorter predicted half-life
- Simple but effective: 45% better predictions than traditional methods

DSR Model (FSRS)

Three-Component Memory Tracking

- Difficulty (D): How hard is this item for you?
- Stability (S): How long until recall drops to 90%?
- Retrievability (R):

$$P(t) = \exp\left(-\frac{t \ln(R_{\text{target}})}{S}\right),$$

- D = Difficulty (0 to ~10) - intrinsic item complexity for YOU
  - Higher D = harder item = slower stability growth
  - Initialized from error count, updated with each review
- S = Stability (in days) - memory strength/durability
  - Time for retrievability to decay from 100% → 90%
  - Grows after successful recalls, shrinks after failures
- R = Retrievability (0 to 1) - current recall probability
- Changes continuously over time (unlike D and S)
- P(t) = Recall probability at time t
- t = Time elapsed since last review (days)
- R_target = Target retention (typically 0.9 = 90%)
- ln = Natural logarithm (mathematical function)
- exp = Exponential function (e^x)

How DSR Works

Step 1 - Initialize each card with three values:

Day 0: Learn "⠁ " (Braille letter A)
→ $D = 5.0$ (medium difficulty, based on first few attempts)
→ $S = 1.0$ (very unstable, brand new memory)
→ $R = 1.0$ (100% - you just learned it)

Step 2 - Retrievability decays over time (continuous):

$R(t) = \exp(-t \cdot \ln(0.9)/S)$

Day 0: $R = \exp(-0 \cdot \ln(0.9)/1.0) = 1.00$ (100%)
Day 1: $R = \exp(-1 \cdot \ln(0.9)/1.0) = 0.90$ (90%)
Day 2: $R = \exp(-2 \cdot \ln(0.9)/1.0) = 0.81$ (81%)
Day 3: $R = \exp(-3 \cdot \ln(0.9)/1.0) = 0.73$ (73%) ← Time to review!

How DSR Works

Step 3 - You review and succeed:

BEFORE review (Day 3):
D = 5.0, S = 1.0, R = 0.73

AFTER successful recall:
→ D decreases slightly: D = 4.8 (item is getting easier for you)
→ S increases significantly: S = 4.2 (memory is now more stable)
→ R resets: R = 1.0 (100% again)

Step 4 - Next review interval is LONGER:

New S = 4.2 days means memory stays strong longer:

Day 3: R = 1.00 (just reviewed)
Day 4: R = 0.98
Day 5: R = 0.96
Day 6: R = 0.94
Day 7: R = 0.92
Day 8: R = 0.90 ← Review again here (after 5 days, not 3!)

How DSR Works

Step 5 - If you FAIL a review:

BEFORE review (Day 8):
D = 4.8, S = 4.2, R = 0.90

AFTER failed recall:
⟶ D increases: D = 5.5 (item is harder than we thought)
⟶ S decreases: S = 2.0 (memory weakened, not as stable)

⟶ R resets: R = 1.0 (but you just relearned it)

The 21 Parameters Control Everything:

How D changes: 4 parameters
How S changes after success: 8 parameters
How S changes after failure: 6 parameters
Initial values: 3 parameters

Total: 21 parameters

Our Solution: Hybrid SM-2+AI

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Core Idea

- Start with SM-2's transparent foundation
- Gradually blend in ML predictions as data accumulates
- Maintain interpretability while gaining adaptability

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The Hybrid Formula

Adaptive Blending Factor (β):

$$\beta(n) = \begin{cases} \beta_{min} = 0.0 & \text{if } n < \beta \\ \min(\beta_{min} + (n - \beta_{threshold}) \cdot \gamma, \beta_{max}) & \text{otherwise} \end{cases}$$

- β (beta) = **Blending weight** (0 to 0.8) - controls how much we trust ML vs SM-2
- n = **Number of reviews** for this specific item
- 0.0 = **β_min** - starting value (pure SM-2)
- 3 = **β_threshold** - minimum reviews before ML kicks in
- 0.1 = **γ** (gamma) - growth rate (how fast we trust ML)
- 0.8 = **β_max** - maximum ML influence (never go to 100%)

The Hybrid Formula

Interval Calculation

$$I\_hybrid = \lfloor (1-\beta) \times I\_SM2 + \beta \times I\_ML \rfloor$$
$$h\_hybrid = (1-\beta) \times h\_SM2 + \beta \times h\_ML$$

- I_hybrid = Final review interval (in days) - when to show this card next
- I_SM2 = SM-2's suggested interval (from ease factor calculation)
- I_ML = ML model's suggested interval (from ensemble prediction)
- h_hybrid = Blended half-life (time until 50% recall probability)
- h_SM2 = SM-2's estimated half-life ≈ I_SM2 / 1.5
- h_ML = ML's predicted half-life (from HLR/DHP/RNN models)
- ⌊ ⌋ = Floor function - round down to nearest whole day
- $(1-\beta)$ = SM-2 weight - decreases as β grows

The Hybrid Formula

Consistency Modulation

$$\beta_{adjusted} = \beta_{base} \cdot \left(0.7 + 0.3 \cdot \frac{\bar{r} - 2}{3}\right)$$

- β_adjusted = Final β after adjustment - may be lower if performance is inconsistent
- β_base = Base β from formula above - before consistency check
- r̄ (r-bar) = Average rating over last 5 reviews (scale 0-5)
- 0.7 = Minimum multiplier - worst case keeps 70% of β
- 0.3 = Variable portion - bonus for consistency
- (r̄ - 2)/3 = Normalized consistency score - ranges from -0.67 to +1.0

Methodology

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Datasets

- FSRS-Anki-20k: 20,000 users, millions of reviews
- Harvard Dataverse: Cross-validation data

CARD_ID	REVIEW_TH	ELAPSED_DAYS	ELAPSED_SECONDS	RATING	STATE	DURATION
0	1	-1	-1	3	0	5737
0	2	0	5	3	1	5700
0	3	4	343517	3	2	60000

HALFLIFE	I	P_RECALL	DELTA_T	R_HISTORY	T_HISTORY	P_HISTORY	D	TOTAL_CNT
1.5	2	0.6299605249474366	1	1	0	0.5000	4	1
1.5	3	0.15749013123685915	4	1,1	0,1	0.5000,0.6300	4	2
6.0	4	0.5	6	1,1,1	0,1,4	0.5000,0.6300,0.1575	4	3

Three Schedulers Compared

- Baseline SM-2: Standard implementation
- Pure-ML: Ensemble of HLR + DHP + RNN models
- Hybrid SM-2+AI: Our proposed approach

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Simulating Human Memory

- **Exponential Forgetting:** Memory decays over time following the formula p = 2^(-t/h), where memories naturally fade without reinforcement, just like real neural connections weaken when not used.
- **Probabilistic Recall:** Memory retrieval isn't perfect - sometimes you remember, sometimes you don't. The simulation uses random outcomes based on memory strength to mimic this real-world variability.
- **Memory Strengthening:** Successful recall makes memories stronger (increases half-life), while failures weaken them (decreases half-life), reflecting how retrieval practice consolidates neural pathways.

Simulating Human Memory

- **Spacing Effect:** Reviews spread over time work better than cramming. The simulation rewards algorithms that space reviews appropriately and penalizes excessive review frequency.
- **Individual Difficulty: S**ome items are naturally harder to learn than others. Each card has unique difficulty (1-10) affecting how quickly it can be mastered, mimicking real learning variability.
- **Cognitive Limits:** The brain can only handle so much per day. A daily budget of 600 units enforces realistic constraints where new learning (12 units) costs more than review (3-9 units), forcing trade-offs between new material and maintenance.

Evaluation Metrics (365-day simulation)

- WTL (Words Total Learned): Items with stable mastery (≥90% recall × 3)
- Daily Cost: Average reviews per day
- Efficiency: Retention per unit effort (SRP ÷ Cost)
- SRP: Sum of recall probabilities across all items
- Recall Curves: Average recall at checkpoints

Key findings

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Day 365 Final Counts

Words Total Learned

Scheduler	Average Item Mastered	vs Pure-ML	vs SM2
SM-2	300	+68.5%	baseline
Hybrid	**305**	**+71.3%**	**+2%**
Pure-ML	178	baseline	-68.5%

Day 365 Final Counts

Reviews Per Day at Day 365

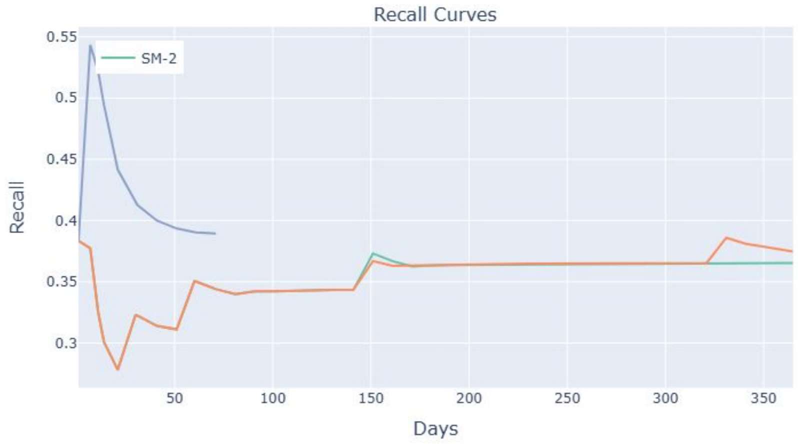
Scheduler	Average Daily Reviews	% of Pure-ML
SM-2	**20.0**	**12% (88% less ↓)**
Hybrid	21.7	13% (87% less ↓)
Pure-ML	166.5	100%

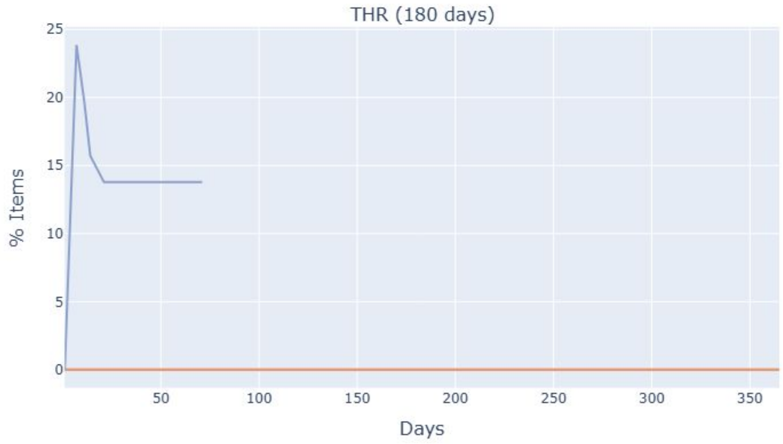
Day 365 Final Counts

Efficiency (SRP ÷ Cost) at Day 365

Scheduler	Average Efficiency	Retention per Review
Pure-ML	**0.389**	**baseline**
Hybrid	0.375	-3.6%
SM-2	0.365	-6.2%

Visualisation





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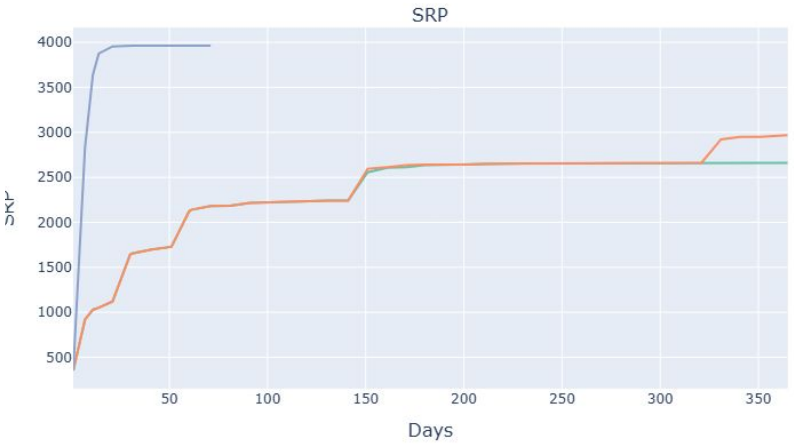

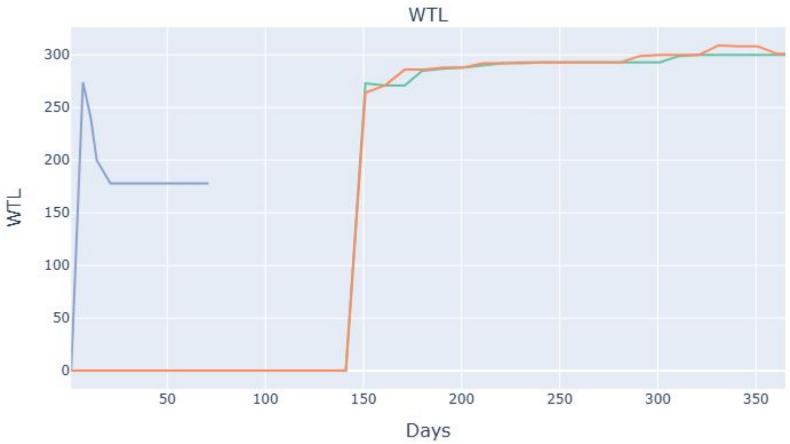

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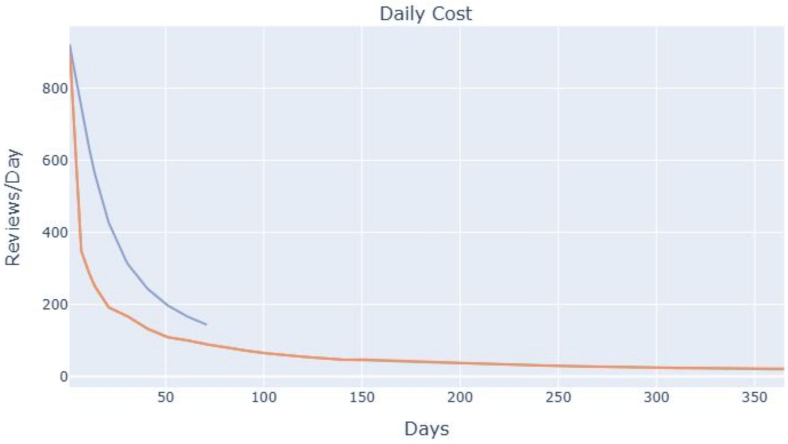

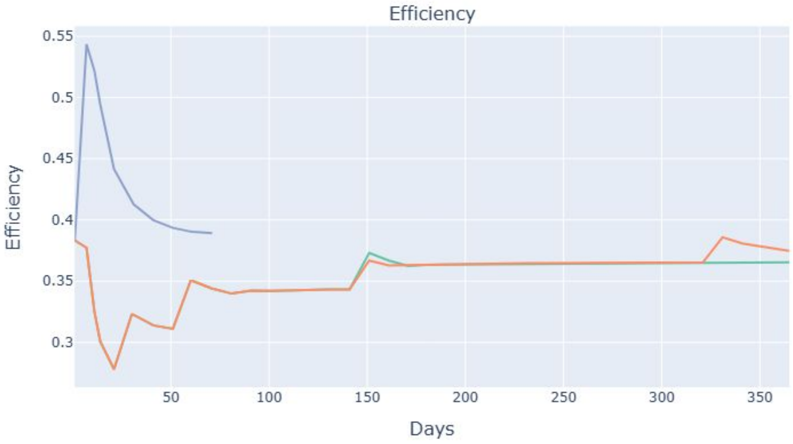

Implementation

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Circuit Design